Jintao Li

Machine Learning and Quantum Computing Engineer, Beijing Academy of Quantum Information Sciences (BAQIS)

Beijing, China · lijt@baqis.ac.cn

Engineer working across machine learning and quantum computing, with first-author papers at ICML 2026 and MICRO 2026 (Best Artifact Honorable Mention). Work spans learning-based combinatorial optimization, reinforcement learning and graph neural networks for quantum compilation, and ML-driven calibration of quantum hardware.

Experience

Engineer, Intelligent Quantum Computing and Simulation Team, Beijing Academy of Quantum Information Sciences (BAQIS), Beijing · Jul 2024 – Present

- First author of LoRe (ICML 2026): adaptive per-step interaction budgets that make learning-based iterative graph solvers faster and more memory-efficient, without retraining the base model.
- First author of QROB (MICRO 2026): a reverse-construction benchmark for quantum compilation that also generates supervision data for learning-based compilers.
- Developed deep reinforcement learning approaches to quantum compilation, using graph neural networks to encode circuit structure for NISQ devices.
- Contributed to ML-based automatic calibration and error suppression for quantum control systems.
- Led front-end development of the Quafu quantum cloud platform, including the circuit editor, data visualization and cloud programming environment.
- Contributing drafter of China's national standardization guidance document "Quantum Computing Service Platform — Interface Requirements" (under review).

Education

- **M.Eng., Electronic Information**, Harbin Engineering University · Sep 2021 – Jun 2024
  - Research: deep reinforcement learning for continuous control
- **B.Eng., Automation**, Harbin Engineering University · Sep 2017 – Jun 2021

Publications

1. **J. Li**, K. Li, R. Wang, Y. Zhao, K. Huang, Y. Wang, J. Zhang, Z.-A. Wang, X. Sun, H. Fan. QROB: Quantifying Realization Overhead in Quantum Compilation via Reverse Construction. *IEEE/ACM MICRO 2026*, pp. 1251–1268. Best Artifact Honorable Mention; artifact on Zenodo.
2. **J. Li**, Y.-Y. Wang, Z.-A. Wang, H. Fan. LoRe: Adaptive Interaction-Evaluation Routing with Per-step Interaction Budgets for Iterative Graph Solvers. *ICML 2026*, PMLR 306, pp. 70969–70982.
3. H. Li, Y. Yang, Y.-H. Shi, Z.-A. Wang, Z. Wang, **J. Li**, et al. Non-equilibrium criticality-enhanced quantum sensing with superconducting qubits. *Science Bulletin* 71(12): 2996–3003, 2026.
4. Z.-A. Wang, Y. J. Wang, J. Zhang, Q. Xu, Y. Zhao, **J. Li**, et al. Hybrid Quantum-Classical Neural Networks for Few-Shot Credit Risk Assessment. *Chinese Physics B*, 2025.
5. X. Zhao, L. He, X. Liu, K. Han, **J. Li**. A novel reinforcement learning framework for optimizing Fixed-Wing UAV flight control strategies. *Aerospace Science and Technology* 166: 110512, 2025.
6. X. Zhao, K. Han, M. Wang, **J. Li**, F. Zong. Ascent Trajectory Planning for Combined-Power Reusable Launch Vehicles Based on the DDPG Algorithm. *2024 China Automation Congress (CAC)*, pp. 2168–2173.

Honors and service

- **Best Artifact Honorable Mention**, MICRO 2026 — one of 3 awards among 94 evaluated artifacts; Available, Reviewed and Reproducible badges.
- **Reviewer**, ACM Transactions on Quantum Computing (2026)
- **Artifact Evaluation Committee**, MICRO 2026

Research projects (participant)

- National Natural Science Foundation of China (NSFC), Joint Fund: Research on a Quantum Cloud Computing Platform Based on Cryogenic Technology, 2026–2029.
- Beijing Natural Science Foundation, General Program: Quantum–Classical Heterogeneous Computing Architecture Design and Quantum Simulation Experiments, 2026–2027.

Patents (China)

- Quantum computing method based on financial models — CN202511331888.1, first inventor
- Optimization method for quantum measurement systems — CN202510080487.7
- Task processing method using quantum cloud computing power — CN202511179017.2
- Method for measuring external-field parameters of superconducting quantum systems — CN202610041442.3
- Method for estimating external-field parameters of quantum systems — CN202610099197.1
- Quantum computer calibration method based on large models and reinforcement learning — CN202610105812.5

Skills

- **Quantum computing:** quantum compilation (qubit routing, lattice-surgery scheduling), fault-tolerant quantum computing, compiler benchmarking, Qiskit
- **Machine learning:** reinforcement learning, graph neural networks, learning-based combinatorial optimization
- **Software:** Python, front-end development for quantum cloud platforms